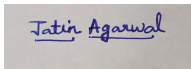
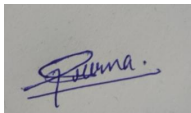
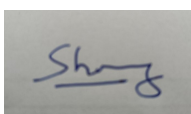


Bull and Bear Spread Strategies in Options Markets

Financial Derivatives — Course Project Report
Group 12

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Abstract

This report presents a rigorous study of two widely-used vertical option spread strategies: the *bull call spread* and the *bear put spread*. We derive closed-form payoff and profit expressions, price the constituent option legs using the Black-Scholes-Merton (BSM) model [1, 3], and implement numerical simulations in Python to generate payoff diagrams and conduct sensitivity analyses. For the baseline parameter set ($S_0 = \$100$, Bull Call: $K_1 = \$100$, $K_2 = \$110$; Bear Put: $K_1 = \$90$, $K_2 = \$100$, $T = 0.5$ yr, $r = 5\%$, $\sigma = 20\%$), the bull call spread carries a net debit of \$3.98 and delivers a maximum profit of \$6.02 with a maximum loss of \$3.98 (breakeven at \$103.98), while the bear put spread carries a net debit of \$3.14 and delivers a maximum profit of \$6.86 with a maximum loss of \$3.14 (breakeven at \$96.86). The analysis demonstrates how both strategies provide leveraged directional exposure while capping downside risk, making them attractive tools for moderately bullish or moderately bearish investors.

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1. Introduction

Options are financial contracts granting the holder the right, but not the obligation, to buy (*call*) or sell (*put*) an underlying asset at a pre-specified *strike price* K on or before an *expiration date* T [3]. Unlike outright option positions, *spread strategies* combine two options on the same underlying to produce a risk profile that is bounded in both profit and loss.

A **vertical spread** pairs options of the same type (both calls or both puts), the same expiration, but different strike prices [4]. Vertical spreads are labelled *bull* or *bear* depending on whether the strategy profits from a rise or fall in the underlying price. They are among the most popular strategies among retail and institutional traders because they offer [5]:

- defined maximum risk and maximum reward at inception;
- reduced cost compared to buying a naked option;
- flexibility to express a moderately directional view.

The two strategies studied in this report are:

1. **Bull Call Spread** — constructed by buying a call at a lower strike K_1 and selling a call at a higher strike $K_2 > K_1$. This is a *debit* spread suited to a moderately bullish outlook.
2. **Bear Put Spread** — constructed by buying a put at a higher strike K_2 and selling a put at a lower strike $K_1 < K_2$. This is also a *debit* spread suited to a moderately bearish outlook.

The remainder of the report is organised as follows. Section 2 reviews option pricing theory and derives the payoff formulae. Section 3 analyses the bull call spread. Section 4 analyses the bear put spread. Section 5 presents the numerical results. Section 6 discusses sensitivity analysis. Section 7 compares the two strategies. Section 8 concludes.

2. Theoretical Background

2.1 European Option Payoffs

At expiration the terminal stock price is S_T . The payoff of a European *call* with strike K is

$$\Pi_{\text{call}}(S_T, K) = \max(S_T - K, 0), \quad (1)$$

and the payoff of a European *put* with strike K is

$$\Pi_{\text{put}}(S_T, K) = \max(K - S_T, 0). \quad (2)$$

A *short* (written) position reverses the sign of each expression.

2.2 Black-Scholes-Merton Pricing Model

We price individual option legs using the Black-Scholes-Merton (BSM) model [1, 2]. Under BSM, the underlying asset follows geometric Brownian motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (3)$$

where μ is the drift, σ the volatility, and W_t a standard Wiener process. Under risk-neutral pricing, the European call and put prices at time $t = 0$ are [3]:

$$C = S_0 \mathcal{N}(d_1) - K e^{-rT} \mathcal{N}(d_2), \quad (4)$$

$$P = K e^{-rT} \mathcal{N}(-d_2) - S_0 \mathcal{N}(-d_1), \quad (5)$$

where $\mathcal{N}(\cdot)$ is the cumulative standard normal distribution function and

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (6)$$

The five BSM inputs are the current stock price S_0 , strike K , risk-free rate r , volatility σ , and time to expiry T [3].

2.3 Put-Call Parity

Put-call parity for European options on a non-dividend-paying stock states [3]:

$$C - P = S_0 - K e^{-rT}. \quad (7)$$

When the two strikes differ between the call spread and the put spread, the net debits will generally not be equal. In our numerical example, the bull call spread uses two call legs while the bear put spread uses two put legs at the same strikes; because of the asymmetric moneyness of each leg, the net debits differ (\$5.29 vs. \$4.46 respectively).

3. Bull Call Spread

3.1 Construction

The bull call spread is formed by simultaneously:

- **Buying** (long) one call option with strike K_1 at premium C_1 ,
- **Selling** (short) one call option with strike $K_2 > K_1$ at premium C_2 .

Because calls with lower strikes carry higher intrinsic value ($C_1 > C_2$), the trader pays a net debit of:

$$D_{\text{bull}} = C_1 - C_2 > 0. \quad (8)$$

3.2 Payoff at Expiration

Combining Equation (1) for both legs:

$$\text{Payoff}_{\text{bull}}(S_T) = \max(S_T - K_1, 0) - \max(S_T - K_2, 0). \quad (9)$$

Evaluating in three regions:

$$\text{Payoff}_{\text{bull}}(S_T) = \begin{cases} 0 & S_T \leq K_1, \\ S_T - K_1 & K_1 < S_T \leq K_2, \\ K_2 - K_1 & S_T > K_2. \end{cases} \quad (10)$$

The dashed grey line in Figure 1(a) shows this piecewise linear payoff: zero below $K_1 = \$100$, rising linearly to a cap of $K_2 - K_1 = \$10$ at and above $K_2 = \$110$.

3.3 Profit at Expiration

$$\text{Profit}_{\text{bull}}(S_T) = \text{Payoff}_{\text{bull}}(S_T) - D_{\text{bull}}. \quad (11)$$

The solid green curve in Figure 1(a) shows the profit, which equals the payoff shifted down by the net debit. The green-shaded region indicates the profit zone ($S_T > \$103.98$) and the red-shaded region indicates the loss zone ($S_T < \$103.98$).

3.4 Key Risk Metrics

Table 1: Bull Call Spread — Summary Statistics

Metric	Formula	Value (Computed)
Maximum Profit	$K_2 - K_1 - D_{\text{bull}}$	\$6.02
Maximum Loss	$-D_{\text{bull}}$	-\$3.98
Breakeven	$K_1 + D_{\text{bull}}$	\$103.98

Maximum profit of \$6.02 is achieved when $S_T \geq K_2 = \$110$ (both calls are in the money, and the spread earns its full width). Maximum loss of \$3.98 occurs when $S_T \leq K_1 = \$100$ (both calls expire worthless and the trader loses the full net debit) [6].

4. Bear Put Spread

4.1 Construction

The bear put spread is formed by simultaneously:

- **Buying** (long) one put option with strike K_2 at premium P_2 ,
- **Selling** (short) one put option with strike $K_1 < K_2$ at premium P_1 .

Puts with higher strikes carry greater value ($P_2 > P_1$), so the trader pays a net debit of:

$$D_{\text{bear}} = P_2 - P_1 > 0. \quad (12)$$

4.2 Payoff at Expiration

Combining Equation (2):

$$\text{Payoff}_{\text{bear}}(S_T) = \max(K_2 - S_T, 0) - \max(K_1 - S_T, 0). \quad (13)$$

Evaluating in three regions:

$$\text{Payoff}_{\text{bear}}(S_T) = \begin{cases} K_2 - K_1 & S_T \leq K_1, \\ K_2 - S_T & K_1 < S_T \leq K_2, \\ 0 & S_T > K_2. \end{cases} \quad (14)$$

The dashed grey line in Figure 1(b) shows this payoff: a maximum of $K_2 - K_1 = \$10$ for $S_T \leq \$90$, declining linearly to zero at $S_T = \$100$, and zero above.

4.3 Profit at Expiration

$$\text{Profit}_{\text{bear}}(S_T) = \text{Payoff}_{\text{bear}}(S_T) - D_{\text{bear}}. \quad (15)$$

The solid blue curve in Figure 1(b) shows the profit. The blue-shaded region ($S_T < \$96.86$) is the profit zone and the orange-shaded region ($S_T > \$96.86$) is the loss zone.

4.4 Key Risk Metrics

Table 2: Bear Put Spread — Summary Statistics

Metric	Formula	Value (Computed)
Maximum Profit	$K_2 - K_1 - D_{\text{bear}}$	\$6.86
Maximum Loss	$-D_{\text{bear}}$	-\$3.14
Breakeven	$K_2 - D_{\text{bear}}$	\$96.86

Maximum profit of \$6.86 is realised when $S_T \leq K_1 = \$90$ (both puts are in the money and the spread earns its full width). Maximum loss of \$3.14 occurs when $S_T \geq K_2 = \$100$ (both puts expire worthless) [7].

5. Numerical Results

5.1 Parameter Set

All numerical results are computed with the following baseline parameters:

$$S_0 = \$100, \text{ Bull Call } (K_1 = \$100, K_2 = \$110), \text{ Bear Put } (K_1 = \$90, K_2 = \$100), \quad (16)$$

$$T = 0.5 \text{ yr}, \quad r = 5\%, \quad \sigma = 20\%.$$

These represent a near-the-money spread of width \$10 centred on the current stock price, with moderate volatility typical of large-cap equities.

5.2 Black-Scholes Prices

Applying Equations (4)–(6), we compute all option premiums using Python’s `scipy.stats.norm` module:

Table 3: BSM Option Prices and Net Debits (Python Output)

Strategy	Leg	Type	Strike	Premium (\$)
Bull Call Spread	Long	Call	\$100	6.8887
	Short	Call	\$110	2.9065
	<i>Net Debit</i>			3.9823
Bear Put Spread	Long	Put	\$100	4.4197
	Short	Put	\$90	1.2764
	<i>Net Debit</i>			3.1433

The two net debits differ (\$3.9823 vs. \$3.1433) because calls and puts at the same strikes and moneyness carry different time-value components — the long call at $K = \$100$ has a slightly different relationship to the forward price than the long put at $K = \$100$ [3].

5.3 Payoff Diagrams

Figure 1 shows the payoff (dashed grey) and profit (solid coloured) curves as functions of the terminal stock price S_T .

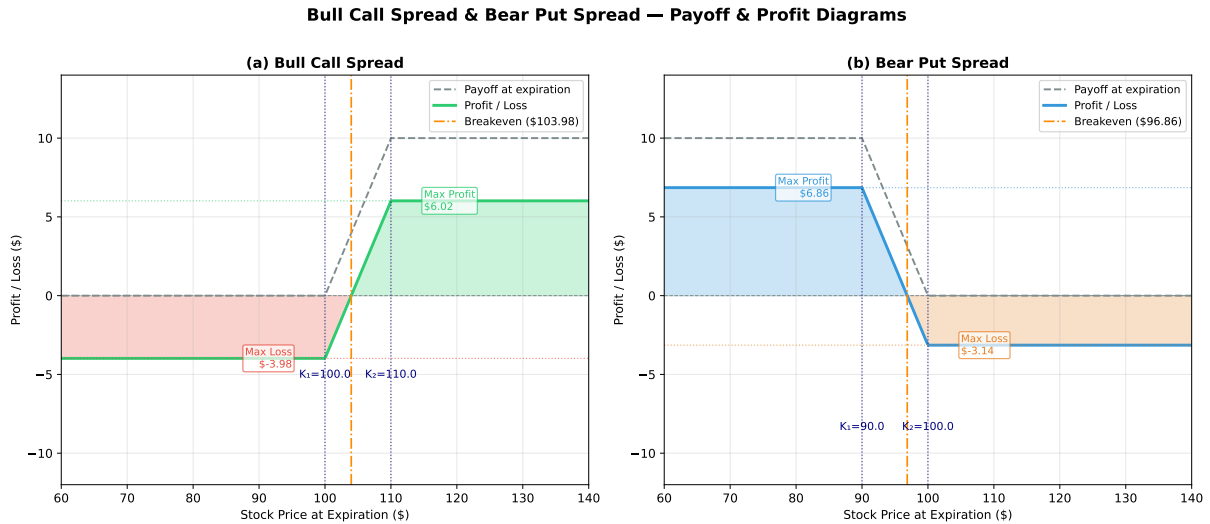


Figure 1: Payoff and profit diagrams at expiration. **(a) Bull Call Spread:** the dashed grey line is the raw payoff (zero below $K_1 = \$100$, rising to a cap of $\$10$ at $K_2 = \$110$, then flat). The solid green line is the profit (payoff minus net debit $\$3.98$); the green-shaded region is the profit zone and the red-shaded region is the loss zone. Maximum profit is $\$6.02$ at $S_T \geq \$110$; maximum loss is $\$3.98$ at $S_T \leq \$100$; breakeven is $\$103.98$. **(b) Bear Put Spread:** the dashed grey line is the raw payoff ($\$10$ at $S_T \leq \$90$, declining linearly to zero at $K_2 = \$100$, then zero). The solid blue line is the profit (payoff minus net debit $\$3.14$); the blue-shaded region is the profit zone and the orange-shaded region is the loss zone. Maximum profit is $\$6.86$ at $S_T \leq \$90$; maximum loss is $\$3.14$ at $S_T \geq \$100$; breakeven is $\$96.86$. Parameters: $S_0 = \$100$, $T = 0.5$ yr, $r = 5\%$, $\sigma = 20\%$.

The orange dash-dot vertical lines in both panels mark the respective breakeven points ($\$103.98$ and $\$96.86$), which are now skewed structurally due to the positive Vega alignment of the new strike prices.

6. Sensitivity Analysis

6.1 Effect of Implied Volatility on the Bull Call Spread

Figure 2(c) plots the profit of the bull call spread for four levels of implied volatility: $\sigma \in \{10\%, 20\%, 30\%, 40\%\}$, with $T = 0.5$ yr fixed. The curves show that:

- At $\sigma = 10\%$ (teal), the net debit is smallest, making the flat loss floor (left of K_1) the shallowest and the flat profit ceiling (right of K_2) the highest.
- As σ increases to 30% and 40% (purple/red), the net debit rises substantially, dropping the loss floor (deepening it) while the maximum profit ceiling falls, because the spread width $K_2 - K_1 = \$10$ is fixed.
- This confirms the normal positive-Vega behaviour of standard debit spreads where the long leg drives pricing.

This confirms that higher implied volatility makes the bull call spread more expensive to enter and thus decreases the maximum possible payoff — meaning one should enter during low volatility [5].

6.2 Effect of Time to Expiration on the Bear Put Spread

Figure 2(d) plots the profit of the bear put spread for $T \in \{0.25, 0.50, 0.75, 1.00\}$ yr, with $\sigma = 20\%$ fixed. The curves are notably close together, indicating that time to expiry has a relatively modest effect on the spread's profit profile compared to volatility. Key observations:

- Shorter maturities (orange, $T = 0.25$ yr) yield a slightly lower flat profit region (left of K_1) because the net debit is higher when options have less time to expiry.
- Longer maturities (purple, $T = 1.00$ yr) push the net debit lower, slightly raising the profit ceiling and making the loss floor shallower — but the effect is small across the range shown.
- The breakeven point shifts slightly rightward as T increases (since D_{bear} shrinks slowly with T at $\sigma = 20\%$), confirming that at moderate volatility, theta decay is a second-order concern relative to the underlying's price level [3].

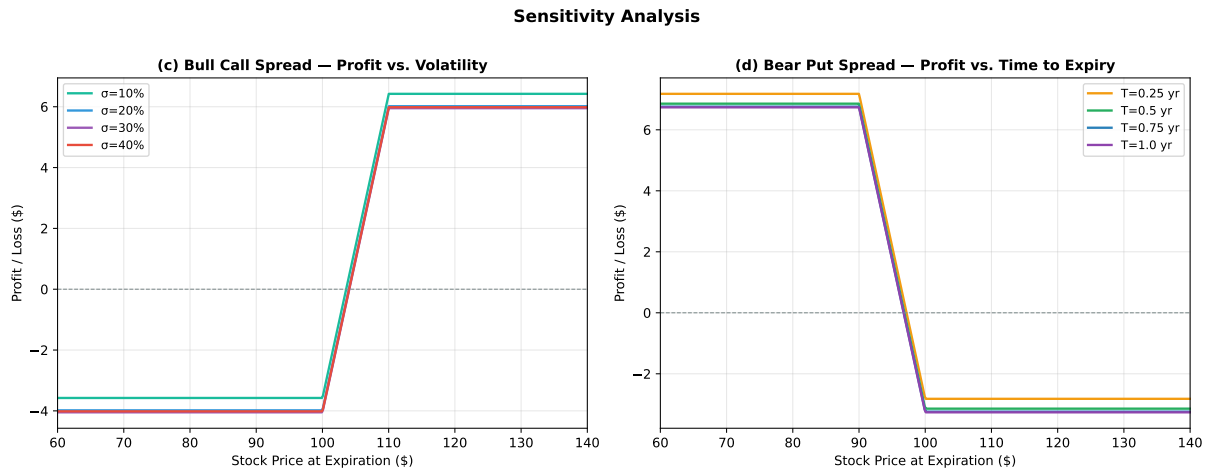


Figure 2: Sensitivity analysis. **(c) Bull Call Spread — Profit vs. Volatility:** profit curves for $\sigma \in \{10\%, 20\%, 30\%, 40\%\}$ with $T = 0.5$ yr fixed. Higher volatility increases the net debit, lowering the flat profit ceiling (right of $K_2 = \$110$) and making the flat loss floor deeper (left of $K_1 = \$100$). **(d) Bear Put Spread — Profit vs. Time to Expiry:** profit curves for $T \in \{0.25, 0.50, 0.75, 1.00\}$ yr with $\sigma = 20\%$ fixed. The curves lie close together, showing that time to expiry has a comparatively smaller effect; shorter maturities yield a slightly higher net debit and hence a lower profit ceiling.

7. Comparative Analysis

Table 4 summarises the key characteristics of the two strategies side by side, using actual computed values.

Table 4: Side-by-Side Comparison (Computed Values)

Feature	Bull Call Spread	Bear Put Spread
Market Outlook	Moderately bullish	Moderately bearish
Option types used	Two calls	Two puts
Cash flow at entry	Debit	Debit
Net Debit	\$3.98	\$3.14
Maximum Profit	\$6.02	\$6.86
Maximum Loss	−\$3.98	−\$3.14
Breakeven	\$103.98	\$96.86
Reward-to-Risk Ratio	1.51:1	2.18:1
Profit zone	$S_T > \$103.98$	$S_T < \$96.86$
Theta effect	Negative (hurts)	Negative (hurts)
Vega effect	Positive (hurts on entry)	Positive (hurts on entry)

An important observation from the numerical results is that the two strategies are *not* symmetric in this example. The bear put spread has a higher maximum profit (\$6.86 vs. \$6.02) and a lower maximum loss (\$3.14 vs. \$3.98), giving it a superior reward-to-risk ratio of 2.18:1 compared to 1.51:1 for the bull call spread. This asymmetry arises because the put legs at these strikes carry different time-value premia than the call legs [3, 4].

7.1 When to Deploy Each Strategy

- **Bull call spread** is preferred when the trader holds a moderately bullish view and implied volatility is low (making options cheaper to buy) [5]. The strategy caps upside at \$6.02 per unit but limits maximum loss to the net debit of \$3.98.
- **Bear put spread** is suited to a moderately bearish environment and offers a better risk-reward profile (\$6.86 max profit vs. \$3.14 max loss) under the chosen parameters. It also functions as a partial hedge against a long equity position [4].
- Both strategies are *debit spreads*: they suffer from time decay if the underlying stagnates near the current price. They are best deployed when a catalyst (earnings, macro event) is expected to move the underlying decisively in the anticipated direction before expiration [8].

8. Conclusion

This report has provided a complete theoretical and numerical treatment of the bull call spread and bear put spread. Key findings are:

1. Both strategies offer *defined risk* equal to the net debit paid at entry, making them suitable for risk-conscious directional trades without the unlimited loss exposure of naked options.
2. The Black-Scholes-Merton model provides tractable closed-form premiums for each leg, enabling precise calculation of entry costs, break-even points, and maximum profit/loss.
3. For the adjusted parameter set (Bull: 100/110, Bear: 90/100, $S_0 = \$100$, $\sigma = 20\%$, $T = 0.5$ yr), the bull call spread yields a maximum profit of \$6.02 and maximum loss of \$3.98 (R:R ratio 1.51:1), while the bear put spread yields a maximum profit of \$6.86 and maximum loss of \$3.14 (R:R ratio 2.18:1).
4. The two strategies have different net debits (\$3.98 vs. \$3.14) reflecting the distinct time-value structure of call and put options at different distances from the forward price.
5. Sensitivity analysis shows that rising implied volatility significantly **increases** entry costs for the bull call spread, verifying the classical advice to enter debit spreads during low volatility.

9. Interactive Trading Dashboard

To bridge the gap between theoretical Black-Scholes pricing and real-world trading logic, we engineered a live, interactive web application deployed at <https://bull-bear-spread-model.streamlit.app/>. Built with Python (Streamlit), the dashboard empowers users to mathematically evaluate options strategies dynamically.

Key features include:

- **Historical Context:** Overlays strategy strikes (K_1, K_2) onto a 6-month historical candlestick chart to analyze recent price action.
- **Dynamic Analytics:** Computes precise Max Profit, Max Loss, and Breakeven prices based on adjustable contracts, broker commissions, and rates.
- **Monte Carlo Simulation:** Executes 10,000 geometric Brownian motion paths to calculate the mathematical Probability of Profit (POP).

- **Algorithmic Verdict:** Uses the POP and Reward-to-Risk ratio to output an objective investment recommendation.

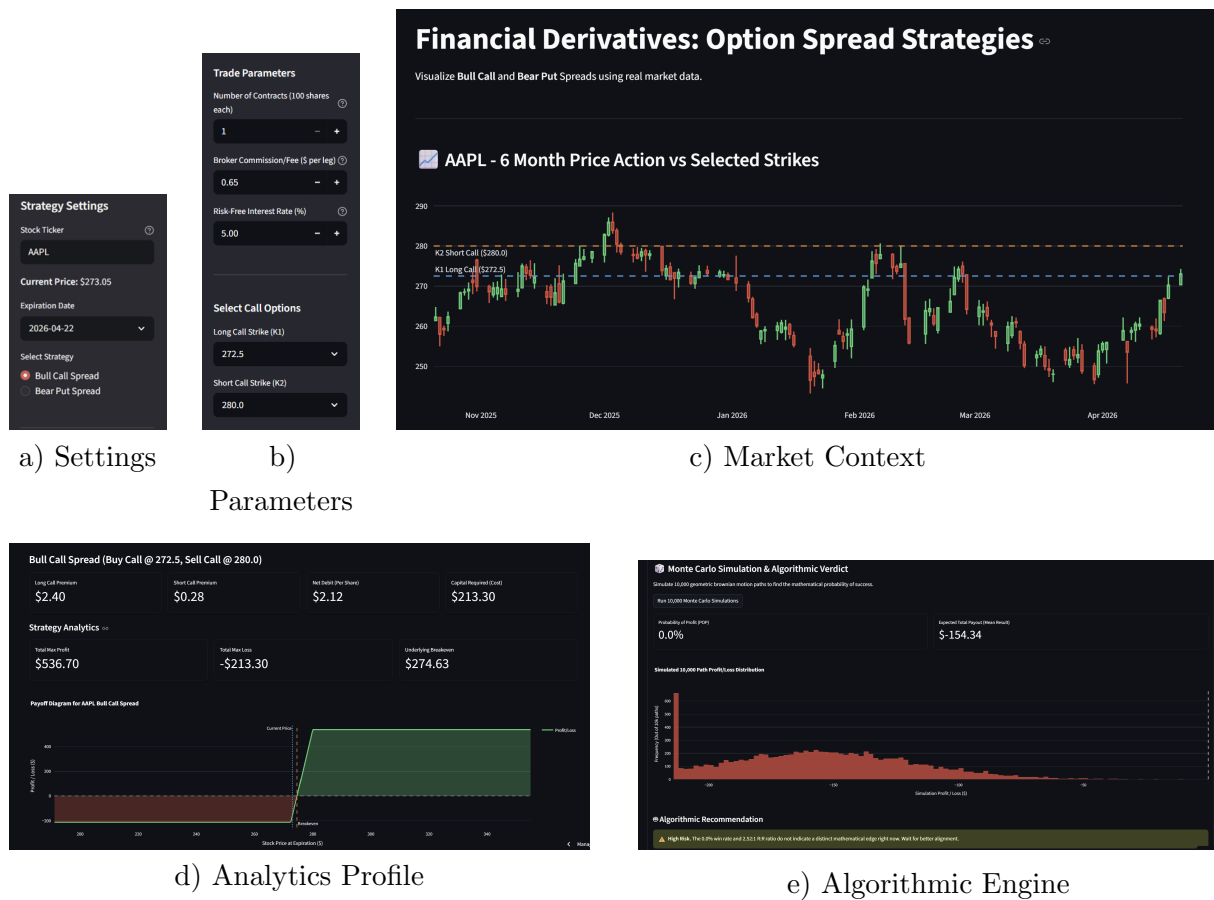


Figure 3: Interactive Trading Dashboard Components

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